

LIM WEN HAO

AI/ML, Agentic AI, Full-stack Web Development, Web Cybersecurity
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PROFESSIONAL SUMMARY

Academically distinguished Year 3 Intelligent Computing majored **JPA-LSPM Scholar (CGPA 3.92)** and **President** of the USM Computer Science Society (2024/2025 academic term) with proven leadership and strong foundational knowledge in **full-stack web development**. Possesses advanced capabilities in **AI/ML engineering**, with deep specialization in **Agentic AI frameworks** and **LLM orchestration**. A proven hackathon and CTF competitor driven to apply rapid prototyping skills and algorithmic optimization to enterprise-grade machine learning challenges.

EDUCATION

Bachelor of Computer Science (Intelligent Computing) with Honours

Universiti Sains Malaysia (USM), Gelugor, Penang

Expected Graduation: October 2027

- **Current CGPA:** 3.92 (until Year 3 Sem 1)
- **Academic Awards:** 5x Top 10s Dean's List Achiever (Year 1 Sem 1 to Year 3 Sem 1)

SKILLS

Programming Languages | Python • JavaScript • C++ • C# • Java

AI/ML & Data Science | LangChain • LangSmith • Agentic AI Frameworks • Model Context Protocol (MCP) • Agent-to-Agent (A2A) • AG-UI • Amazon Bedrock • TensorFlow • Keras • OpenCV • Pandas • NumPy

Web Development | HTML • CSS • JavaScript • ReactJS • Astro • Vercel • Tailwind CSS

Cloud & Backend Services | AWS • AWS Lambda • Node.js • Firebase • Supabase • Cloudflare

Databases | MySQL • PostgreSQL • SQLite • MongoDB • Airtable

Mobile Development | Flutter

Tools & Version Control | Git • GitHub • Bitbucket • Docker • Anaconda • Figma • Adobe XD • Copilot • Jira

CI/CD & Deployment | GitHub Actions • Jenkins • CloudBees

Soft Skills | Team player • Strategic leadership • Effective communication • Deliver results

* Note: For a detailed skill lists, refer to my [GitHub Profile](#).

HONORS & AWARDS

JPA-LSPM Scholar (2022–2027) | Awarded for academic excellence (SPM: 9A+ 1A).

Regional Champion (Malay Region), Singapore AI Safety Red Teaming Challenge (2026): Achieved the highest score in the region for exploiting vulnerabilities, evaluating data leakage risks, and testing multicultural AI safety frameworks.

rawSEC NERACA 3.0 CTF x CYRES USM (2025) | Fourth place out of entire Northern Region of Malaysia.

Top 20 Finalist, Great Malaysia AI Hackathon (2025) | Tech Lead for MyGov Hub (out of 350+ teams) in ASEAN's Largest On-Site AI Hackathon.

Top 8% World Wide CTF (WWCTF) (2025) | Ranked among 1600+ global teams, demonstrating proficiency in cybersecurity and exploitation challenges.

LEADERSHIP EXPERIENCE

President, Computer Science Society (CSS) USM

Universiti Sains Malaysia (USM), Gelugor, Penang

July 2024 – July 2025

- **Executive Management:** Directed the strategic planning and daily operations of the largest departmental tech society at USM, coordinating a 60+ member team and managing the annual calendar of all technical and non-technical events.
- **Flagship Project Direction:** Served as Project Director for critical initiatives, including the Varsity Hackathon 2025 and the PayNet & NACSA Industrial Tour 2025; also organized the technical Embedded LLM AI Workshop.
- **Corporate Engagement & Outreach:** Cultivated university partnerships with 30+ industries to secure sponsorship, resources, and facilitate corporate engagement; speaker for university outreach (UOWM-KDU Penang); official representative at major tech events (e.g., DevFest, GitHub Field Day).

WORK EXPERIENCE

Kyodo Engineering Sdn. Bhd. | React • TypeScript • Tailwind CSS • Supabase • PostgreSQL • Optimization Algorithms
Freelance Software Developer *December 2025 – Current (Expected: September 2026)*

- **Inventory Workflow Optimization:** Architected and deployed a highly customized full-stack web application tailored to streamline how steel designers manage and catalog raw structural materials.
- **Algorithmic Waste Minimization:** Engineered an optimization engine using the Knapsack Greedy Algorithm to calculate high-efficiency steel pipe cutting patterns, drastically cutting material waste during fabrication.
- **Database & Frontend Integration:** Developed a highly interactive, responsive layout using modern frontend practices linked to a structured PostgreSQL relational database for real-time inventory adjustments.

Keysight Technologies Sdn. Bhd. | LangChain • LangSmith • MCP • A2A • AG-UI • Agentic AI • Python • Docker
Full-time AI/ML Intern *March 2026 – August 2026*

- **Agentic AI Architecture:** Designed and implemented advanced AI agents leveraging the Agentic AI framework, Model Context Protocol (MCP), and Agent-to-Agent (A2A) communications.
- **LLM Orchestration & Monitoring:** Utilized LangChain for complex multi-agent orchestration and integrated LangSmith to systematically trace, evaluate, and optimize production LLM pipelines.
- **Production Deployment:** Collaborated in engineering robust internal AI solutions, prioritizing high-efficiency prompting strategies, workflow integration, and rigorous latency testing.

Telebort Technologies Sdn. Bhd. | Python • YOLO • Machine Learning • HTML • CSS • Scratch
Part-time Technical Instructor *December 2023 – May 2024*

- **Demonstrated Effective Communication:** Taught and mentored 52 students (ages 8-16) across 110+ hours, simplifying complex concepts in AI with Python and Data Analysis.
- **Achieved High Student Satisfaction:** Delivered 53 lessons and maintained exceptional average ratings (4.94/5.0 lesson and 4.91/5.0 instructor), with over 90% of students reporting clear understanding.
- **Curriculum Coverage:** Specialized in delivering foundational programs including AI with Python, Data Analysis, Web Design (HTML/CSS), and creative computing (Scratch).

ViTrox Corporation Bhd. | Machine Learning • Data Preparation • Data Annotation
Part-time ML Data Annotator *April 2022 – July 2022*

- **Data Annotation:** Contributed to the model training data preparation pipeline for Machine Learning (ML) and Machine Vision Inspection systems.
- **High-Volume Execution:** Accurately labeled over 100 images per day (including complex circuit boards and document samples) over three months.
- **AI Pipeline Exposure:** Demonstrated foundational experience in the critical data annotation phase, building robust datasets for supervised learning models.

PROJECTS

MyGov Hub – AI-Powered Government Services Assistant | LLM • AWS Bedrock • AWS Lambda • MongoDB • Python
Technical Lead *September 2025 – October 2025*

<https://github.com/MyGovHub-Goodbye-World>

- **Recognition:** Top 20 Finalist at the largest ASEAN on-site hackathon, Great Malaysia AI Hackathon 2025 (AWS Malaysia).
- **Architecture & AI Integration:** Led the development and deployment of a secure conversational AI chatbot, leveraging **AWS Bedrock** and the **Amazon Nova LLM** to unify fragmented government e-services (licenses, payments, summons).
- **Backend & Data Integration:** Designed and deployed scalable services using **AWS Lambda** and **MongoDB** to manage data, integrate payment APIs (Billplz), and ensure high-availability system performance.

Seismic Bump Prediction (ML/DL Research) | Python • TensorFlow • Keras • Pandas • Scikit-learn
Academic Research Project *March 2025 – July 2025*

<https://github.com/WenHao1223/seismic-bump-prediction>

- **Advanced Modeling:** Conducted a two-phase research project to predict seismic hazards, developing and benchmarking baseline models (KNN, DTC, SVM) for comparative analysis.

- **Deep Learning Integration:** Engineered a **Neural Network (NN)** model to achieve superior performance, specifically optimizing for high **Recall** in detecting high-energy seismic bumps.
- **Data & Model Optimization:** Applied **GridSearchCV** for robust hyperparameter tuning and leveraged Python libraries (Pandas, Scikit-learn) for end-to-end data processing and model deployment.

PIXEL USM – Project Innovation & eXploration Website | ReactJS • Astro • Tailwind CSS • Firebase • Airtable • Cloudflare • GCP

Head of Technical Department (2024) / Associate Head (2025)

May 2024 – July 2025

<https://pixelusm.com/>

- **Strategic Leadership:** Led a technical team across two project cycles, applying **agile principles** to coordinate technical teams and manage deployment timelines in high-pressure, event-driven environments.
- **Full-Stack Development:** Directed the full-stack development and deployment of the official event website, overseeing the implementation of **React/Astro** for interactive front-end experiences.
- **Backend & Cloud Integration:** Engineered a secure and scalable backend by integrating **Firebase** (Auth, Cloud Functions, Database) for application logic, **Airtable** for content management, and **Google Cloud Platform (GCP)** services, prioritizing stability and security features like ReCAPTCHA.

** Note: For a detailed project lists, refer to my [LinkedIn Project](#) page and [GitHub Repositories](#).*

LICENSES & CERTIFICATES

Foundation: Building Reliable Agents | Issued by LangChain • March 2026

Project: Deep Agents | Issued by LangChain • March 2026

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | Issued by Oracle • October 2025 (Expires October 2027)

Foundational C# with Microsoft | Issued by freeCodeCamp • October 2025

MongoDB Node.js Developer Path | Issued by MongoDB • April 2025

Malaysian University English Test (MUET) | Band 4.5 • Issued by MPM Malaysia • June 2023

** Note: For a detailed license and certificate lists, refer to my [LinkedIn Licenses & Certificates](#) page.*